

SHT-40

TEMPERATURE AND HUMIDITY SENSOR

Datasheet

The SHT40 sensor module measures humidity ($\pm 1.8\%$ RH) and temperature ($\pm 0.2\text{ }^{\circ}\text{C}$) via I²C, operating from 0–100 %RH and -40 – $125\text{ }^{\circ}\text{C}$. It runs on 3.3–5 V with 0.4 μA power. NIST-traceable and JEDEC-qualified, it’s ideal for weather monitoring, HVAC, agriculture, and IoT, with mobile app cloud integration.

Key Features

- High-accuracy digital humidity and temperature sensing using I²C interface.
- Ultra-low power operation, ideal for IoT and portable devices.
- Reliable performance even in condensing environments.
- Integrated variable power heater for moisture management.



SHT40-AD1B-R2 Sensor Board

Technical Specifications

Parameter	Value
Relative Humidity Accuracy	$\pm 1.8\%$ RH (typical)
Temperature Accuracy	$\pm 0.2\text{ }^{\circ}\text{C}$ (typical)
Supply Voltage (Breakout)	3.3 V to 5 V
Average Current Consumption	0.4 μA (at 1 Hz measurement rate)
Idle Current	80 nA
Communication Interface	I ² C (Fast Mode Plus, CRC checksum)
Operating Range	0 to 100 %RH, $-40\text{ }^{\circ}\text{C}$ to $+125\text{ }^{\circ}\text{C}$
Condensing Environments	Fully functional in condensing environments
Heater	Integrated variable power heater
Default I ² C Address	0x44

Applications

- Weather monitoring systems
- HVAC (Heating, Ventilation, and Air Conditioning) systems
- Smart agriculture and greenhouse control
- IoT and smart home devices
- Industrial and consumer environmental monitoring